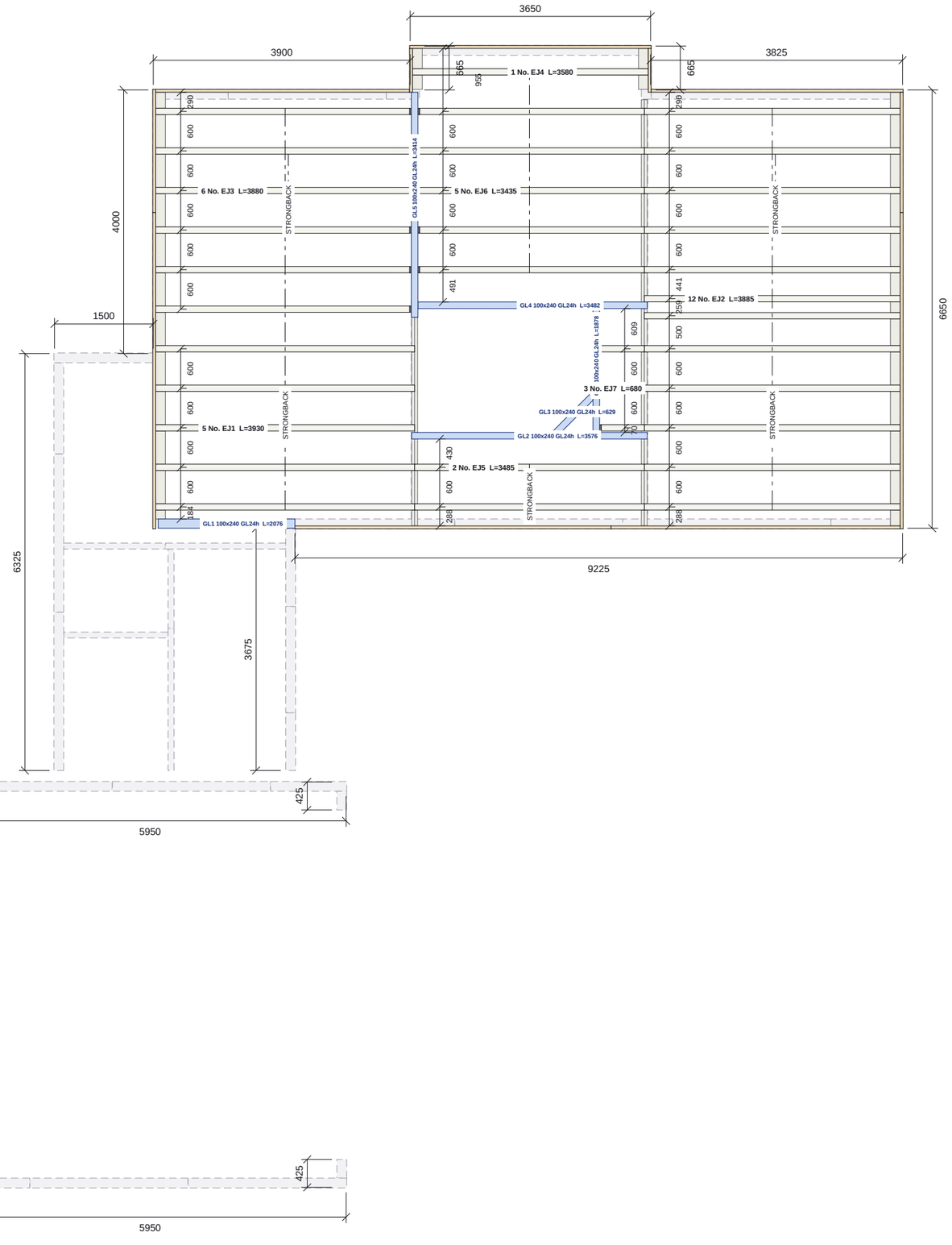



## FIRST FLOOR STRUCTURE PLAN - INDICATIVE

Scale 1:75 @ A3 | Ground floor walls below shown grey dashed; joists by Acorn Timber Engineering

0 1 2 3 4 5m

## FIRST FLOOR STRUCTURE SCHEDULES

### BEAMS

| Mark | Section | Length | No. |
|---|---|---|---|
| GL1 | 100x240 GL24h | 2076 | 1 |
| GL2 | 100x240 GL24h | 3576 | 1 |
| GL3 | 100x240 GL24h | 629 | 1 |
| GL4 | 100x240 GL24h | 3482 | 1 |
| GL5 | 100x240 GL24h | 3414 | 1 |
| GL6 | 100x240 GL24h | 1878 | 1 |
| PK | 14mm marine ply packer, 100 wide | 15.1m | 6 |

PK: 14mm marine ply fixed to the top of each 240mm glulam to give the full 254mm floor depth (240 + 14 = 254, flush with the joists and edgebinder). Glulam and steel supplied by TLS; set out to faces as dimensioned; bearings to engineer. Steel TOS above underside of sole plate.

### METAL-WEB JOISTS - BY ACORN

| Type | Section | Length | No. |
|---|---|---|---|
| EJ1 | 97x254 metal web | 3930 | 5 |
| EJ2 | 97x254 metal web | 3885 | 12 |
| EJ3 | 97x254 metal web | 3880 | 6 |
| EJ4 | 97x254 metal web | 3580 | 1 |
| EJ5 | 97x254 metal web | 3485 | 2 |
| EJ6 | 97x254 metal web | 3435 | 5 |
| EJ7 | 97x254 metal web | 680 | 3 |

Acorn Timber Engineering supply, with 14 hangers, 5 strongback runs and noggins. Not in TLS procurement. Indicative; Acorn's layout supersedes.

### EDGEBINDER, WAIST BAND, BREATHER

| Item | Quantity | Order |
|---|---|---|
| Edgebinder 254 (45x195 + 45x59 rip) | 35.2m | 9 x 4.8m 45x195 |
| 45x59 C24 rips, 3 per 45x195 | 35.2m | 3 x 4.8m 45x195 |
| 45x195 C24 total | | 12 x 4.8m |
| OSB/3 waist band 290x9 | 15 strips | 5 x 1197x2397 |
| Breather 450 x 2.7m strips, 150 lap | 14 strips | 15 (1 roll) |

Run = modelled edgebinders. Timber cut-optimised +5%; OSB +7.5%; membrane laps +5%.

### FLOOR DECKING

| Item | Area | Order |
|---|---|---|
| 22mm Egger P5 600x2400 | 73.6 m2 | 55 sheets |
| D4 adhesive 1L (per 6.5 sheets) | | 9 x 1L |

Laid area from the model (27 full sheets + 49 cut pieces) = 51.1 sheets, +7.5%.

### NOTES

1. INDICATIVE ONLY. Metal-web joists, hangers, strongbacks and noggins are designed and supplied by Acorn Timber Engineering; this drawing is superseded by Acorn's engineered floor layout on receipt.
2. Glulam beams 100x240 GL24h (dark blue) set out to the faces dimensioned, bearing on multi-ply studs to engineer's drawing 26-431-S-0010. 14mm marine ply fixed to the top of each 240mm glulam to give the full 254mm floor depth. Steel UB254x102x28 S355 supplied by TLS.
3. Edgebinder 254mm = 45x195 C24 with a 45x59 C24 rip fixed on top (rips cut 3 per 45x195, 3mm kerf).
4. OSB/3 waist band 290x9 to the outer face of the edgebinder, strips ripped 4 per 1197x2397 sheet.
5. Breather membrane over the waist band in 450 deep x 2.7m strips, each lapped 150 over the next.
6. 22mm Egger P5 flooring glued with D4 adhesive, fixed to Acorn's specification. Dimensions to frame and beam faces.

### REVISIONS

| REV | DATE | DESCRIPTION |
|---|---|---|
| C01 | 14.09.26 | Construction issue |

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

| CLIENT | PROJECT |
|---|---|
| MP Adams | MP Adams Plot 4, Eckington |

DRAWING TITLE

**FIRST FLOOR STRUCTURE - INDICATIVE**

Beams, set-out and metal-web joist positions; Acorn layout supersedes

| SCALE | SIZE | DATE | DRAWN |
|---|---|---|---|
| 1:75 | A3 | 14.09.26 | TS |

| STATUS | DRAWING No. | REV |
|---|---|---|
| CONSTRUCTION | P4-FL-01 | C01 |

Do not scale. All dimensions in mm. Check all dimensions on site.